

IAS PRELIMS/MAINS

Ecology & Environment

A stylized illustration of a green city skyline with various skyscrapers, wind turbines, and a sun, set against a light green background with a globe-like shape.

A COMPREHENSIVE COVERAGE

MODULE - 41

CLIMATE CHANGE - 2

- Climate change has the potential to tip out of balance the relatively stable climate in which human civilization has been built and jeopardize the security of water and food.
- The current year carbon dioxide concentration reached a global average of 405 parts per million, which is the highest ever recorded, matched only by the climate which existed 800,000 years ago.
- The top 20 warmest years on record have come since 1990.



IMPACTS OF CLIMATE CHANGE

1 PRECIPITATION PATTERN SHIFTS - DROUGHTS AND FLOODS

- Climate change disrupts the water cycle and precipitation.
- According to scientists from the Intergovernmental Panel on Climate Change (IPCC), there is increasing probability for more intense droughts and precipitation events. A pattern might emerge of lengthy dry spells interspersed with brief but heavy precipitation and possible flooding.
- Erratic rainfall patterns lead to drought. Also climate change can aggravate drought.
 - ✓ For example, rising temperatures lead to increased evaporation rates and plant transpiration, which results in water loss in soil and plants.

ipcc
INTERGOVERNMENTAL PANEL ON
climate change



IMPACTS OF CLIMATE CHANGE



2

MELTING GLACIERS

- Called 'water towers,' mountains are source to numerous rivers and other freshwater sources.
- Mountain meltwater and runoff provide more than 50 percent of the world's freshwater.
- As global temperatures elevate, mountain glaciers and snow are melting at an unprecedented rate.
- Many mountain glaciers are in retreat. If glaciers melt away, they can't be restored. Thus, areas that previously depended on glaciers for freshwater will then have to seek other sources.



2 MELTING GLACIERS

THE RETREAT OF GLACIERS SINCE 1850

- The retreat of glaciers since 1850 affects the availability of fresh water for irrigation, domestic use, animals and plants that depend on glacier-melt, and, in the longer term, increases the level of the oceans.
- The temporal coincidence of glacier retreat with the measured increase of atmospheric greenhouse gases is often cited as a key evidence of global warming.
- Mountain ranges such as the Himalayas, Alps, Rocky Mountains, Andes, as well as some isolated mountains such as Mount Kilimanjaro in Africa, are showing some of the largest proportionate glacial losses.



2

MELTING GLACIERS

SNOW DROUGHT

- Further complicating mountain water storage capability is the greater likelihood that warmer temperatures make precipitation fall as rain rather than snow, which is called “snow drought.”
- Snow drought leads to reduced water availability.
- When snow and ice collect on mountaintops, the snowmelt releases water slowly – to streams, rivers and reservoirs – throughout the spring and summer.
- On the other hand, when rain falls on mountaintops, water runs off and quickly fills reservoirs to capacity so the excess runoff water can’t be stored.
- Because rain flows faster than melting snow, levels of soil moisture and groundwater recharge may be reduced.



CLIMATE CHANGE

IMPACTS OF CLIMATE CHANGE

3 CLIMATE CHANGE AND POLAR REGIONS

- Average air temperatures in the Arctic region have increased by about 5°C over the last 100 years. Recent data shows that there will be almost no summer sea ice cover left in the Arctic in the next few decades.
- The Antarctic ice sheet is the largest single mass of ice on earth, accounting for around 90% of all fresh water on the earth's surface and spanning almost 14 million sq km.
- This ice plays an important role in influencing the world's climate, reflecting back the sun's energy and helping to regulate global temperatures.
- Parts of the west Antarctica are warming rapidly. Even small-scale melting is likely to have significant effects on global sea level rise.



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